



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

DECOMPOSER CIRCULATION PUMPS (P-2302A/B)

Docu. No. : 14780-8120-PS-P2302
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 6

TITLE :

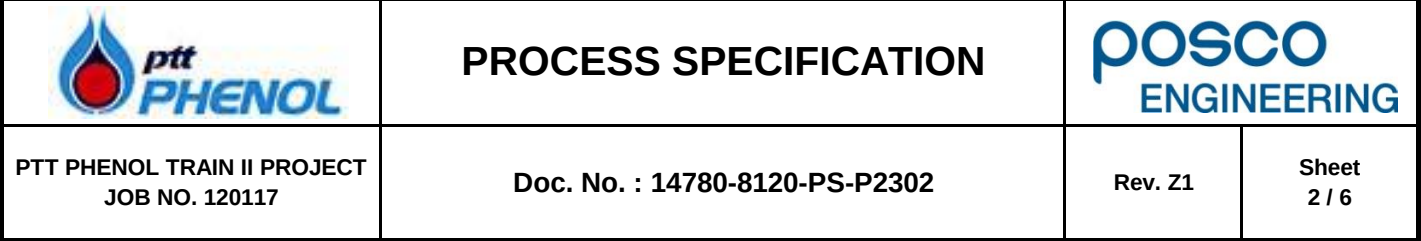
DECOMPOSER CIRCULATION PUMPS

(P-2302A/B)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-P2302
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	FEB 25, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUL 30, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREP'N	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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Record of Revision

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CENTRIFUGAL PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 501

SHEET 3 / 6

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	25-Apr-12	MW	MEH				

REV													
1	APPLICABLE TO: ☐ PROPOSALS ☐ PURCHASE ☒ AS BUILT												
2	FOR _____ UNIT _____												
3	SITE _____ SERVICE Decomposer Circulation Pump												
4	NO. REQ 2 PUMP SIZE _____ TYPE Between Bearing NO. STAGES 1												
5	MANUFACTURER _____ MODEL _____ SERIAL NO. _____												
6	NOTES: INFORMATION BELOW TO BE COMPLETED: ☐ BY PURCHASER ☐ BY MANUFACTURER ☒ BY MANUFACTURER OR PURCHASER												
7	GENERAL												
8	PUMPS TO OPERATE IN (PARALLEL) NO. MOTOR DRIVEN 2 NO. TURBINE DRIVEN _____												
9	(SERIES) WITH _____ PUMP ITEM NO. P-2302A/B PUMP ITEM NO. _____												
10	GEAR ITEM NO. _____ MOTOR ITEM NO. _____ TURBINE ITEM NO. _____												
11	GEAR PROVIDED BY _____ MOTOR PROVIDED BY _____ TURBINE PROVIDED BY _____												
12	GEAR MOUNTED BY _____ MOTOR MOUNTED BY _____ TURBINE MOUNTED BY _____												
13	GEAR DATA SHT. NO. _____ MOTOR DATA SHT. NO. _____ TURBINE DATA SHT. NO. _____												
14	OPERATING CONDITIONS (5.1.3)												
15	☐ CAPACITY, NORMAL 2,020 m³/h RATED 2,020 m³/h												
16	OTHER _____ m³/h												
17	☐ SUCT PRESS MAX./RATED 11.60 / 1.234 kg/cm²[g]												
18	☐ DISCHARGE PRESSURE 4.640 kg/cm²[g]												
19	☐ DIFFERENTIAL PRESSURE 3.41 kg/cm²												
20	☐ DIFF. HEAD 37.6 m NPSHA 5.61 m												
21	☐ PROCESS VARIATIONS _____												
22	☐ STARTING CONDITIONS _____												
23	SERVICE: ☒ CONT. ☐ INTERMITTENT (STARTS/DAY) _____												
24	☐ PARALLEL OPERATION REQ'D												
25	SITE AND UTILITY DATA												
26	LOCATION: ☐ INDOOR ☐ HEATED ☐ UNDER ROOF												
27	☒ OUTDOOR ☒ UNHEATED ☐ PARTIAL SIDES												
28	☒ GRADE ☐ MEZZANINE _____												
29	ELECTRIC AREA CLASSIFICATION												
30	CL 1 GR IIA DIV ZONE 2												
31	☐ WINTERIZATION REQ'D ☒ TROPICALIZATION REQ'D.												
32	SEE NOTES ON SHEET 6 *												
33	SITE DATA												
34	ALTITUDE _____ BAROMETER _____												
35	RANGE OF AMBIENT TEMPS: MIN/MAX. _____ / _____												
36	RELATIVE HUMIDITY: MIN / MAX _____ / _____ %												
37	UNUSUAL CONDITIONS: ☐ DUST ☐ FUMES												
38	OTHER Tropical, Marine												
39	UTILITY CONDITIONS:												
40	STEAM: DRIVERS HEATING												
41	MIN _____												
42	MAX _____												
43	ELECTRICITY DRIVERS HEATING CONTROL SHUTDOWN												
44	VOLTAGE _____												
45	HERTZ _____												
46	PHASE _____												
47	COOLING WATER:												
48	TEMP. INLET _____ MAX. RETURN _____												
49	PRESS. NORM _____ DESIGN _____												
50	MIN RETURN _____ MAX ALLOW DP _____												
	FOULING FACTOR _____												



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REV	DATE	BY	APVD	REV	DATE	BY	APVD
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REV	CONSTRUCTION	CONSTRUCTION (CONT)																						
	APPLICABLE STANDARD: ☒ API 610 11TH EDITION ☐ OTHER _____ (SEE REMARKS)	☐ SHAFT DIAMETER BETWEEN BEARINGS _____ mm ☐ SPAN BETWEEN BEARING CENTERS _____ mm ☐ SPAN BETWEEN BEARING & IMPELLER _____ mm	1																					
	PUMP TYPE: <table style="width:100%;"> <tr> <td>☐ OH2</td> <td>☐ BB1</td> <td>☐ VS1</td> <td>☐ VS6</td> </tr> <tr> <td>☐ OH3</td> <td>☒ BB2</td> <td>☐ VS2</td> <td>☐ VS7</td> </tr> <tr> <td>☐ OH4</td> <td>☐ BB3</td> <td>☐ VS3</td> <td>☐ OTHER</td> </tr> <tr> <td>☐ OH5</td> <td>☐ BB4</td> <td>☐ VS4</td> <td></td> </tr> <tr> <td>☐ OH6</td> <td>☐ BB5</td> <td>☐ VS5</td> <td></td> </tr> </table>	☐ OH2	☐ BB1	☐ VS1	☐ VS6	☐ OH3	☒ BB2	☐ VS2	☐ VS7	☐ OH4	☐ BB3	☐ VS3	☐ OTHER	☐ OH5	☐ BB4	☐ VS4		☐ OH6	☐ BB5	☐ VS5		REMARKS: COUPLINGS: DRIVER - PUMP ☐ MAKE _____ ☒ MODEL _____ ☐ RATING _____ ☐ LUBRICATION _____ ☒ LIMITED END FLOAT REQUIRED _____ ☒ SPACER LENGTH _____ mm ☒ SERVICE FACTOR _____	2	
☐ OH2	☐ BB1	☐ VS1	☐ VS6																					
☐ OH3	☒ BB2	☐ VS2	☐ VS7																					
☐ OH4	☐ BB3	☐ VS3	☐ OTHER																					
☐ OH5	☐ BB4	☐ VS4																						
☐ OH6	☐ BB5	☐ VS5																						
	NOZZLE CONDITIONS: <table border="1" style="width:100%; border-collapse: collapse;"> <thead> <tr> <th>SIZE (IN)</th> <th>FLANGE CLASS</th> <th>FAC'G</th> <th>POSITION</th> </tr> </thead> <tbody> <tr> <td>SUCTION</td> <td>300</td> <td>RF</td> <td></td> </tr> <tr> <td>DISCHARGE</td> <td>300</td> <td>RF</td> <td></td> </tr> <tr> <td>BALANCE DRUM</td> <td></td> <td></td> <td></td> </tr> </tbody> </table>	SIZE (IN)	FLANGE CLASS	FAC'G	POSITION	SUCTION	300	RF		DISCHARGE	300	RF		BALANCE DRUM				DRIVER HALF COUPLING MOUNTED BY: ☐ PUMP MFR. ☐ DRIVER MFR. ☐ PURCHASER ☐ COUPLING PER API 671	3					
SIZE (IN)	FLANGE CLASS	FAC'G	POSITION																					
SUCTION	300	RF																						
DISCHARGE	300	RF																						
BALANCE DRUM																								
	PRESSURE CASING CONNECTIONS: <table border="1" style="width:100%; border-collapse: collapse;"> <thead> <tr> <th>NO.</th> <th>SIZE (NPS)</th> <th>TYPE</th> </tr> </thead> <tbody> <tr> <td>☐ DRAIN</td> <td></td> <td>Provide With Valve</td> </tr> <tr> <td>☐ VENT</td> <td></td> <td></td> </tr> <tr> <td>☐ PRESSURE GAUGE</td> <td></td> <td></td> </tr> <tr> <td>☐ TEMP GAUGE</td> <td></td> <td></td> </tr> <tr> <td>☐ WARM-UP</td> <td></td> <td></td> </tr> <tr> <td>☐ BALANCE / LEAK-OFF</td> <td></td> <td></td> </tr> </tbody> </table>	NO.	SIZE (NPS)	TYPE	☐ DRAIN		Provide With Valve	☐ VENT			☐ PRESSURE GAUGE			☐ TEMP GAUGE			☐ WARM-UP			☐ BALANCE / LEAK-OFF			BASEPLATES: ☐ API BASEPLATE NUMBER _____ (APPENDIX M) ☐ NON-GROUT CONSTRUCTION:	4
NO.	SIZE (NPS)	TYPE																						
☐ DRAIN		Provide With Valve																						
☐ VENT																								
☐ PRESSURE GAUGE																								
☐ TEMP GAUGE																								
☐ WARM-UP																								
☐ BALANCE / LEAK-OFF																								
	☐ CYLINDRICAL THREADS REQUIRED	REMARKS: MATERIAL ☐ APPENDIX H CLASS _____ A-8 ☐ MIN DESIGN METAL TEMP _____ 15 °C ☐ BARREL/CASE 316 SS IMPELLER 316 SS ☐ CASE/IMPELLER WEAR RINGS Hard-Faced 316 SS ☐ SHAFT 316 SS ☐ DIFFUSERS _____ ☐ COUPLING SPACER/HUBS _____ ☐ COUPLING DIAPHRAGMS (DISKS) _____	5																					
	CASING MOUNTING: (SEE SEPARATE SHEET FOR VERTICALS) ☒ CENTERLINE ☐ NEAR CENTERLINE ☐ FOOT ☐ SEPARATE MOUNTING PLATE ☐ IN-LINE		6																					
	CASING SPLIT: ☐ AXIAL ☒ RADIAL		7																					
	CASING TYPE: ☐ SINGLE VOLUTE ☒ MULTIPLE VOLUTE ☐ DIFFUSER ☐ OVERHUNG ☒ BETWEEN BEARINGS ☐ BARREL		8																					
	CASE PRESSURE RATING: ☐ MAX ALLOWABLE WORKING PRESSURE _____ kg/cm ² [g] @ _____ 0 ☐ HYDROTEST PRESSURE _____ kg/cm ² [g] ☐ SUCT'N PRESS. REGIONS MUST BE DESIGNED FOR MAWP		9																					
	ROTATION: (VIEWED FROM COUPLING END) ☐ CW ☐ CCW ☐ IMPELLERS INDIVIDUALLY SECURED		10																					
	REMARKS: ☐ BOLT OH3 PUMP TO PAD/FOUNDATION SHAFT: ☐ SHAFT DIAMETER AT COUPLING _____ mm		11																					
	REMARKS: SUCTION TYPE Double IMPELLER TYPE Closed Pumps shall comply with UOP Standard Specifications 5-11 and 7-12 See Project Specification 533.		12																					
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		BEARINGS AND LUBRICATION BEARING (TYPE/NUMBER): ☐ RADIAL Ball / _____ ☐ THRUST Ball / _____ ☐ REVIEW AND APPROVE THRUST BEARING SIZE LUBRICATION: ☒ GREASE ☒ FLOOD ☐ RING OIL ☒ FLINGER ☐ PURGE OIL MIST ☒ PURE OIL MIST ☐ CONSTANT LEVEL OILER PREFERENCE (SEE REMARKS) ☐ PRESSURE LUBE SYS ☐ API-610 ☐ API-614 ☒ OIL VISC. ISO GRADE _____	38																					
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CENTRIFUGAL PUMPS AND DRIVERS

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				

REV	BEARINGS AND LUBRICATION (CONT)	MECHANICAL SEAL OR PACKING (CONT)	
	☒ OIL HEATER REQ'D ☐ ELECTRIC ☐ STEAM ☐ OIL PRESS TO BE GREATER THAN COOLANT PRESS REMARKS 	☐ VAPOR PRESSURE _____ kg/cm ² [a] @ _____ °C ☐ HAZARDOUS ☐ FLAMMABLE ☐ OTHER _____ ☐ FLOW RATE MAX/MIN _____ / _____ m ³ /h ☐ PRESSURE REQUIRED MAX/MIN _____ / _____ kg/cm ² [g] ☒ TEMPERATURE REQUIRED MAX/MIN _____ / _____ °C	1
	MECHANICAL SEAL OR PACKING SEAL DATA: ☐ SEE ATTACHED API-682 DATA SHEET ☐ NON-API 682 SEAL ☐ APPENDIX H SEAL CODE _____ ☒ SEAL MANUFACTURER _____ ☒ SIZE AND TYPE _____ / _____ Pressurized Dual ☒ MANUFACTURER CODE _____ SEAL CHAMBER DATA: ☒ TEMPERATURE _____ ☒ PRESSURE _____ kg/cm ² [g] ☒ FLOW _____ m ³ /h ☐ SEAL CHAMBER SIZE (TABLE 2.3) _____ ☐ TOTAL LENGTH _____ mm ☐ CLEAR LENGTH _____ mm SEAL CONSTRUCTION: ☐ SLEEVE MATERIAL 316 Austenitic Stainless Steel ☐ GLAND MATERIAL 316 Austenitic Stainless Steel ☐ AUX SEAL DEVICE Non-Sparking Floating Throttle Bushing * ☒ JACKET REQUIRED GLAND TAPS: ☒ FLUSH (F) ☒ DRAIN (D) ☒ BARRIER (B) ☒ QUENCH (Q) ☒ COOLING (C) ☒ LUBRICATION (G) ☒ HEATING (H) ☒ LEAKAGE ☒ PUMPED FLUID (P) ☒ BAL'NCE FLUID (E) ☒ EXTERNAL FLUID INJECTION (X) SEAL FLUIDS REQUIREM'T AND AVAILABLE FLUSH LIQUID: NOTE: IF FLUSH LIQUID IS PUMPAGE LIQUID (AS IN FLUSH PIPING PLANS 11 TO 41), FOLLOWING FLUSH LIQUID DATA IS NOT REQ'D. ☐ SUPPLY TEMPERATURE MAX/MIN _____ / _____ °C ☐ RELATIVE DENSITY (SPECIFIC GRAVITY) _____ @ _____ °C ☐ NAME OF FLUID _____ ☐ SPECIFIC HEAT _____ kcal/(kg.°C) ☐ VAPOR PRESSURE _____ kg/cm ² [a] @ _____ °C ☐ HAZARDOUS ☐ FLAMMABLE ☐ OTHER _____ ☐ FLOW RATE MAX/MIN _____ / _____ m ³ /h ☐ PRESSURE REQUIRED MAX/MIN _____ / _____ kg/cm ² [g] ☐ TEMPERATURE REQUIRED MAX/MIN _____ / _____ BARRIER FLUID: ☐ SUPPLY TEMPERATURE MAX/MIN _____ / _____ °C ☐ RELATIVE DENSITY (SPECIFIC GRAVITY) _____ @ _____ °C ☐ NAME OF FLUID Silicone	QUENCH FLUID: ☐ NAME OF FLUID _____ ☐ FLOW RATE _____ m ³ /h SEAL FLUSH PIPING: ☐ SEAL FLUSH PIPING PLAN _____ API Plan 11 ☒ TUBING ☐ CARBON STEEL ☐ PIPE ☒ STAINLESS STEEL ☐ AUXILIARY FLUSH PLAN _____ API Plan 53B ☒ TUBING ☐ CARBON STEEL ☐ PIPE ☒ STAINLESS STEEL ☐ PIPING ASSEMBLY: ☒ THREADED ☒ UNIONS ☐ SOCKET WELDED ☐ FLANGED ☒ TUBE TYPE FITTINGS ☒ LOW PRESSURE SWITCH (PLAN 52/53) TYPE _____ ☒ PRESSURE GAUGE (PLAN 52/53) ☒ LOW LEVEL SWITCH (PLAN 52/53) TYPE _____ ☒ LEVEL GAUGE (PLAN 52/53) ☐ TEMP INDICATOR (PLANS 21, 22, 23, 32, 41) ☒ HEAT EXCHANGER (PLAN 52/53) REMARKS	2
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CENTRIFUGAL PUMPS AND DRIVERS

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1	13-Aug-12	MW	DRL				

REV

NOTES:

* SEE BASIC ENGINEERING DESIGN QUESTIONNAIRE (BEDQ), ENGINEERING DESIGN INFORMATION (EDI) PROJECT 963763 A.4, SECTIONS 3 AND 4.

The following notes apply to those services for which they are specified in this Project Specification:

PUMP NOTES:

2. Pump vendor shall provide non-contacting pressurized dry gas seals. The barrier gas shall be Nitrogen from the plant Nitrogen header available at 7 kg/cm²(g). Pump vendor shall provide a seal control panel for each pump. The panel shall include the following:

- Coalescing filter (removal efficiency of 98% on liquids 1 micron or greater and solids 3 microns or greater)
- Pressure regulating valve
- Pressure gauge
- Flow meter low range
- Flow meter high range

Alarm switches:

- Low Pressure
- High Flow

The alarm switches shall activate a control center trouble alarm (by others). The contractor and pump vendor shall coordinate switch selection.

A mechanical amplifier shall be provided by the seal vendor to boost the nitrogen buffer supply pressure when the normal nitrogen supply pressure does not exceed either:

- the pump suction pressure by 2.1 kg/cm², or
- the pump maximum suction pressure by 0.35 kg/cm² psi.

When these conditions exist, the seal gas control panel scope shall be expanded to include the amplifier and a minimum 2 gallon stainless steel accumulator vessel with local pressure gauge.

6. Mechanical seal parts and packings shall be suitable for handling aromatic hydrocarbons.

9. The "capacity, other" is based on 50% of the lowest normal operating flow.

11. Pumps shall be mounted on cover plates to be furnished by vessel supplier to suit sump tank opening. Each pump shall include a stainless steel suction screen. A steel discharge pipe shall extend through the cover plate terminating in a Cl. 300 flanged connection. Contractor to determine depth of setting. Pump will be spared by common warehouse spare pump P-1304S.

42. VOC packing valves should be supplied.

C1. Casing volume to be minimized and subjected to critical volume restriction

C2. Cumene Hydroperoxide is thermally unstable that can initiate self-heating reaction at elevated temperatures

C3. Acid Aromatics sump pump P-2304A shall be identical to the existing Acid Aromatics sump pump in PPCL Unit 1 (Item No. P-1304A/S). Existing warehouse spare pump P-1304S shall be used as a common spare for both services.

MOTOR NOTES:

M5. Non Overloading at all points on pump curve

M10. Motor shall have torque capability to accelerate the pump to operating speed with discharge shutoff valve open, and the discharge check valve closed against a static pressure of 1.8 kg/cm² (g)

M12. Motor bearings shall be lubricated with a pure (dry sump) oil mist system.

M13. The standby pump motor shall be connected to an alternate independent power supply.

GENERAL NOTES:

G1. Lubrication system is applied to high voltage motor's and all pumps' bearings except for vertical and diaphragm pumps.

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